

Stefan Stepanović

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SUMMARY

Computer vision research engineer with several years of experience in industrial R&D, working on structure from motion, neural rendering and 3D generative modeling. Focused on connecting classical geometric vision and deep learning approaches to advance 3D scene understanding and its applications.

EXPERIENCE

YDrive AI

Senior Research Engineer in Computer Vision

Nov 2021 – Present

Generative AI approaches to 3D:

- Leading YDrive's efforts in creating systems capable of generating and understanding unbounded 3D scenes.
- In charge of research and development of novel 3D generative models based on Gaussian Splatting.

Structure from motion:

- Designed and maintained reliable and efficient state-of-the-art SFM solution for large scale reconstructions.
- Delivered advancements in bundle adjustment through GPU efficient implementations and partitioning strategies. Reduced reconstruction time of thousands of images from days to hours.
- Integrated deep learning into classical SFM for faster and more accurate reconstructions. Adapted and extended academic works for pose estimation and camera calibration for production grade deployment.

Neural rendering:

- Developed point cloud-based neural rendering system using neural network as rasterization model, achieving real-time 30FPS HD rendering with high visual fidelity.
- Introduced spatio-temporal training regularization methods for mitigating flickering artifacts.
- Led the productization of a highly optimized solution based on Unreal Engine plugin and WebRTC streaming integration for 5G enabled mobile devices.

Internship management:

- Designed assignments and interviewed two generations of interns.
- Defined research projects aligned with state of the art academic advancements and company road-map.
- Supervised multiple internships with projects spanning from advanced techniques for point cloud rasterization to geometric regularization for Gaussian Splatting.

École polytechnique fédérale de Lausanne (EPFL)

July 2021 – Nov 2021

Research fellow at Visual Intelligence Lab under Prof. Amir Zamir

Representation learning on mid-level tasks in computer vision:

- Researched task balancing methods through loss weighting that led to stable training and comparable results across all tasks.
- Designed dropout regularization methods to enforce hierarchy in latent representations through PCA-like properties. This enabled coarse-to-fine control of modalities directly at representation level.

YDrive AI

Computer Vision Engineering Intern

Sept 2019 – Dec 2019
July 2020 – Apr 2021

Structure From Motion:

- Optimized mapping pipeline for generating high-definition 3D maps reducing quadratic complexity to linear.
- Researched and delivered point-free SFM solution using general 3D curves enabling reconstruction even on texture poor areas.

EDUCATION

Bachelor of Science

2018 – 2022

University of Belgrade, Faculty of Electrical Engineering

CGPA – 9.95/10.00

Dept. of **Signals & Systems**

Thesis: Reducing complexity of bundle adjustment through partitioning methods.

VOLUNTEER ACADEMIC WORK

PSIML (Practical Seminar in Machine Learning)

2022 – Present

Mentor

Mentorship to undergraduate and graduate students providing a strong foundational education in machine learning through lectures and practical work.

Examples of practical projects include writing and training NeRF and image diffusion models from scratch.

PFE (Robotics education camp)

2019 – Present

Mentor

Lecturer and mentor for motivated high school students in technical fields. The purpose of the camps is education in various fields, from applied physics to machine learning and control systems.

TALKS

Unreal Day - Epic Games

Fall 2023

Delivered a presentation to 400+ attendees to demonstrate YDrive's neural rendering advancements and their impact on the VFX and gaming industry.

PERSONAL PROJECTS

Sentence Completion using DSSM

Summer 2019

Repurposed and trained Deep Structured Semantic Models for next word prediction.

Done as a project during Practical Seminar in Machine Learning.

Automatic Extractive Summarization of Multiple Documents

Summer 2018

Reimplemented work from paper on recursive neural network approach to text ranking and its application to extractive summarization.

Single Camera Plane Estimation and Tracking

Summer 2017

Structure from motion pipeline from scratch in C++ simplified to planar scenes.

Implemented core algorithms and structures for modeling multiple-view geometry.

Application for database and product management for Local Business

Fall 2018

Developed a management application specialized for local business (Drvogalant Ltd.) in C# and maintained it since.

SKILLS

Languages

English (C2 proficiency), Serbian (mother tongue)

Programming languages

Python, C++, CUDA, C

Industry knowledge

Git, Bash, CMake, Linux, Docker

Toolset

Pytorch, Ceres, Eigen, Tensorflow, OpenGL